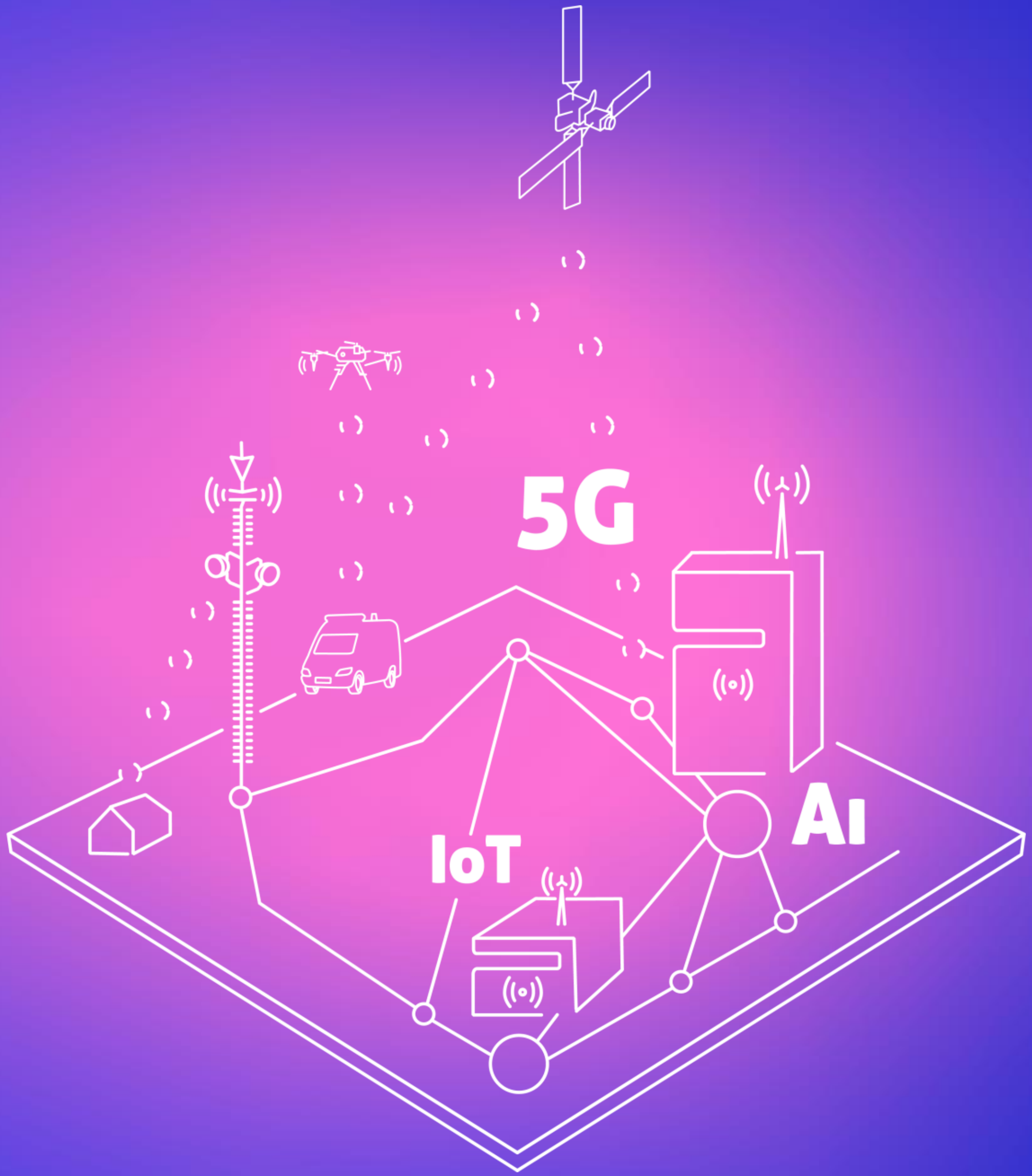
CELLULAR EXPERT

6. Importing data



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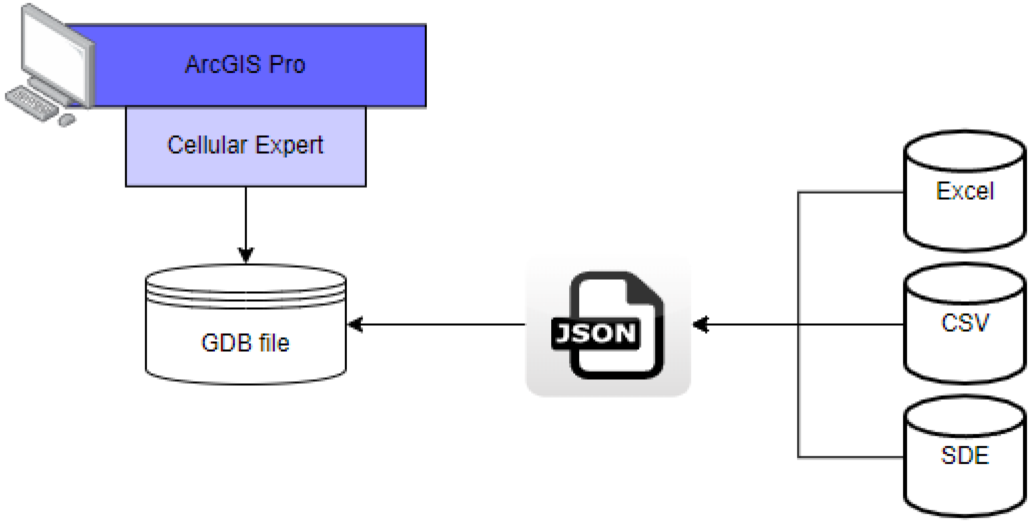
Network import for CE for ArcGIS Pro

- From:
  - Excel
  - CSV
  - SDE table

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Cell Name	Height Above Ground, m	Azimuth	Tilt	Frequency	Cell Powe	Gain	Bandwidtl	TxMIMO	RxMIMO	Tech	Tower Na	latitude	longitude	Duplex
2	NBa 01	33	86	0	3500	40	16.32	100	4	4	5G	1	54.72501	25.22863	TDD
3	NBa 02	30	357	0	3500	40	16.32	100	4	4	5G	1	54.72525	25.22862	TDD
4	NBa 03	30	247	0	3500	40	16.32	100	4	4	5G	1	54.72501	25.22863	TDD
5	NBb 01	30	305	0	3500	40	16.32	100	4	4	5G	2	54.7367	25.26736	TDD
6	NBb 02	30	99	0	3500	40	16.32	100	4	4	5G	2	54.7367	25.26736	TDD
7	NBb 03	30	212	0	3500	40	16.32	100	4	4	5G	2	54.7367	25.26736	TDD



- To Cellular Expert Workspace:
  - gdb database

Network objects

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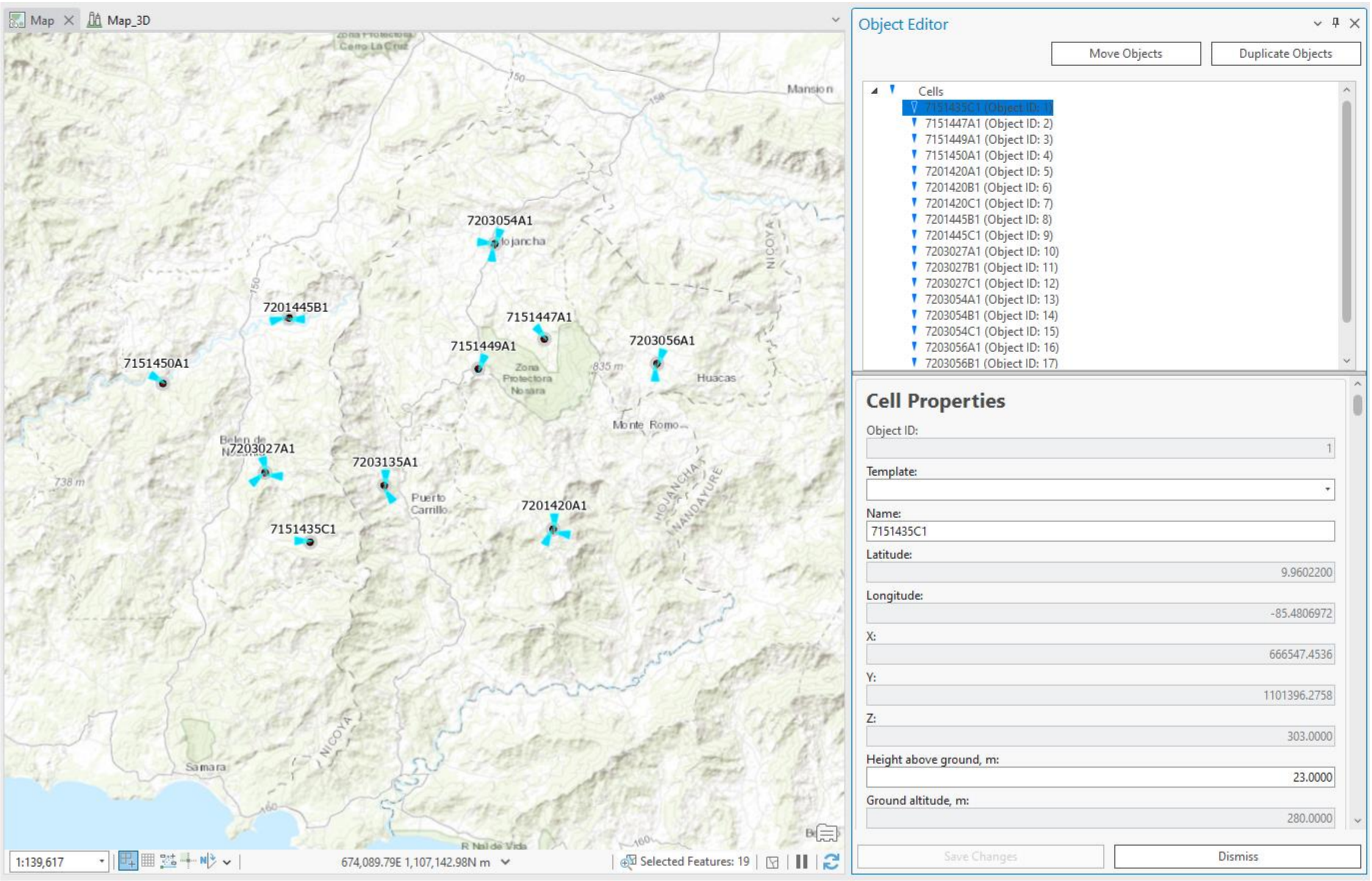
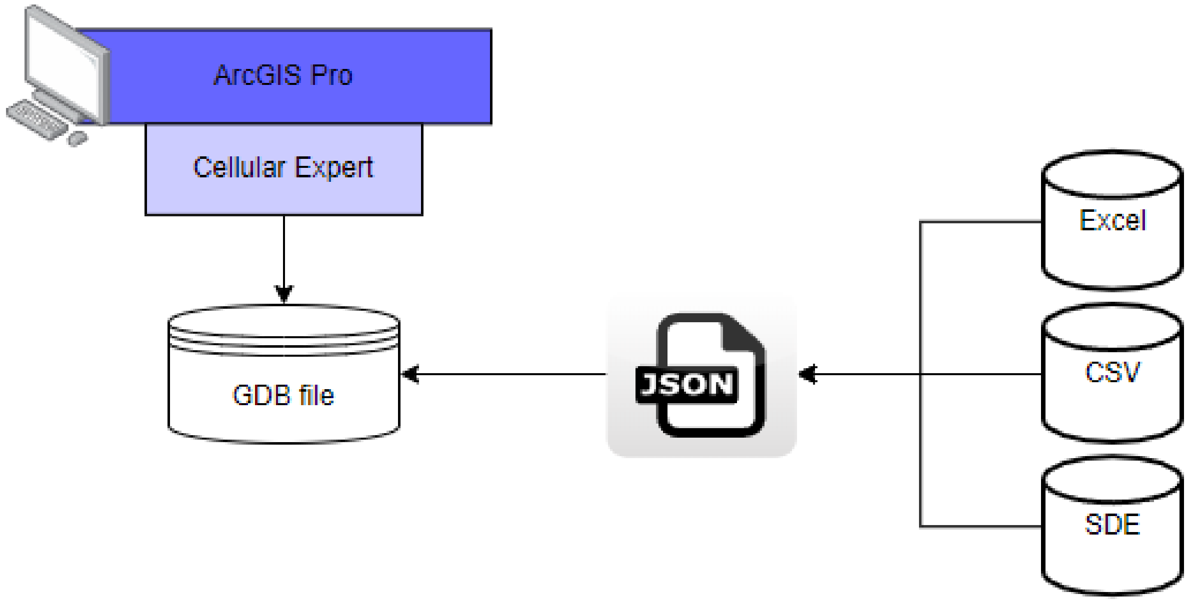
- Cells
  - Sites
- Sites (if siteid parameter is defined)



Mapping file

- Json type file, can be edited with Notepad



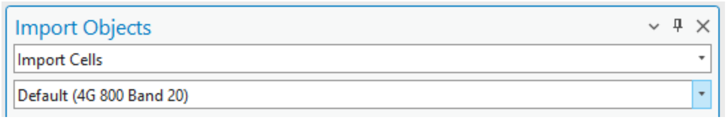

```
{
        "table_name": "Cells",

        "fields": [
                {
                        "current_name": "Cell Name",
                        "destination_name": "cell_name",
                        "default_value": ""
                },
                {
                        "current_name": "latitude",
                        "destination_name": "latitude",
                        "default_value": ""
                },
                {
                        "current_name": "longitude",
                        "destination_name": "longitude",
                        "default_value": ""
                },
                {
                        "current_name": "Height Above Ground",
                        "destination_name": "height",
                        "default_value": ""
                },
                {
                        "current_name": "Azimuth",
                        "destination_name": "azimuth",
                        "default_value": ""
                },
                {
                        "current_name": "Tilt",
                        "destination_name": "tilt",
                        "default_value": ""
                },
                {
                        "current_name": "Frequency",
                        "destination_name": "frequency",
                        "default_value": ""
                },
                {
                        "current_name": "Cell Power",
                        "destination_name": "power",
                        "default_value": ""
                },
                {
                        "current_name": "",
                        "destination_name": "misc_loss",
                        "default_value": ""
                },
```

Mapping file structure

- **“current_name”** - name of the value that is written in the data file. As an example “freq_mhz” is a column name in the data file and will be changed to “frequency” when the mapping file is applied and objects are imported.

```
"current_name": "",
"destination_name": "cell_name",
"default value": ""
```



- **“destination_name”** - the proper name of the property (table column name) in the Cellular

- **“default_value”** – The default value applies when an object in the data file lacks a specific property. The same value will be applied for all imported objects.

Cells: generate Cell Name

- Check option: Generate Cell Name
  - Latitude
  - Longitude
  - Azimuth
  - Frequency
  - Power
  - Height
  - Antenna gain

Apply prediction model

Select Territory Polygon (Optional)

- Polygon type feature class/shape file
- ModelID and ConfigID is a must
- Option appears when Import HCM patterns option is active.

Parameters for Cell object

cell_name – cell identifier (recommended unique value).

latitude - Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

longitude - Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

height – Cell height above the terrain.

azimuth - Cell direction from the North in degrees.

tilt - Mechanical tilt value.

frequency - Frequency value in MHz.

power - Power value in dBm.

antenna_gain - Antenna gain value from the applied antenna.

misc_loss - Miscellaneous loss value in dB.

Parameters for Cell object

bandwidth - Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

noise_figure - Value in dB. Required for 4G and 5G technologies.

downlink_duplex_factor - Value range from 0 to 1. Required for 4G and 5G technologies, and used for Downlink Throughput calculations.

subcarrier_spacing - Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

tx_mimo - Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

rx_mimo - Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

active_antenna_effect - The parameter is dedicated to smart antenna modelling. The default value is 0, but if massive MIMO is used, a smart antenna effect can be included to lower the interference and boost throughput. Recommended

values:

For MIMO 32x32 – value 6.

For MIMO 64x64 – value 9.

Parameters for Cell object

cell_load - The parameter is described in percentages and varies from 0 to 100. It describes how the cell is loaded. The Cell load affects RSSI, RSRQ, DL Throughput calculations. For example, if the Cell load is higher, the DL Throughput is lower.

technology - Describes the technology of cell. Possible values are 2G, 3G, 4G, 5G and WiFi.

prediction_model_id – prediction model identification:

If value 1 – ITU-R P.452

If value 2 – UniMacro

If value 3 – CEC ITU-R

If value 4 – LOS ITU-R P.525

If value 5 – ITU-R P.368

prediction_model_configuration_id – prediction model configuration identification in define prediction model (prediction_model_id value).

frequency_group – helps to manage different frequency groups, RF prediction will do separate prediction for different frequency_group value automatically.

antenna_id- antenna identification in the database.

duplex_mode - Required for 4G and 5G technologies, possible values FDD or TDD.

site_id – to automatically create Site object for cells, define site name field here. Must be a text format.

Exercise

Description: C:\CE_Course\0. Descriptions

Name: 6. Importing data.pdf

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Thank you!

Tel.: +370 5 2150575

Email: info@cellular-expert.com

S.Konarskio g. 28A LT-03127 Vilnius
Lithuania

www.cellular-expert.com